

Architecture and Design Document

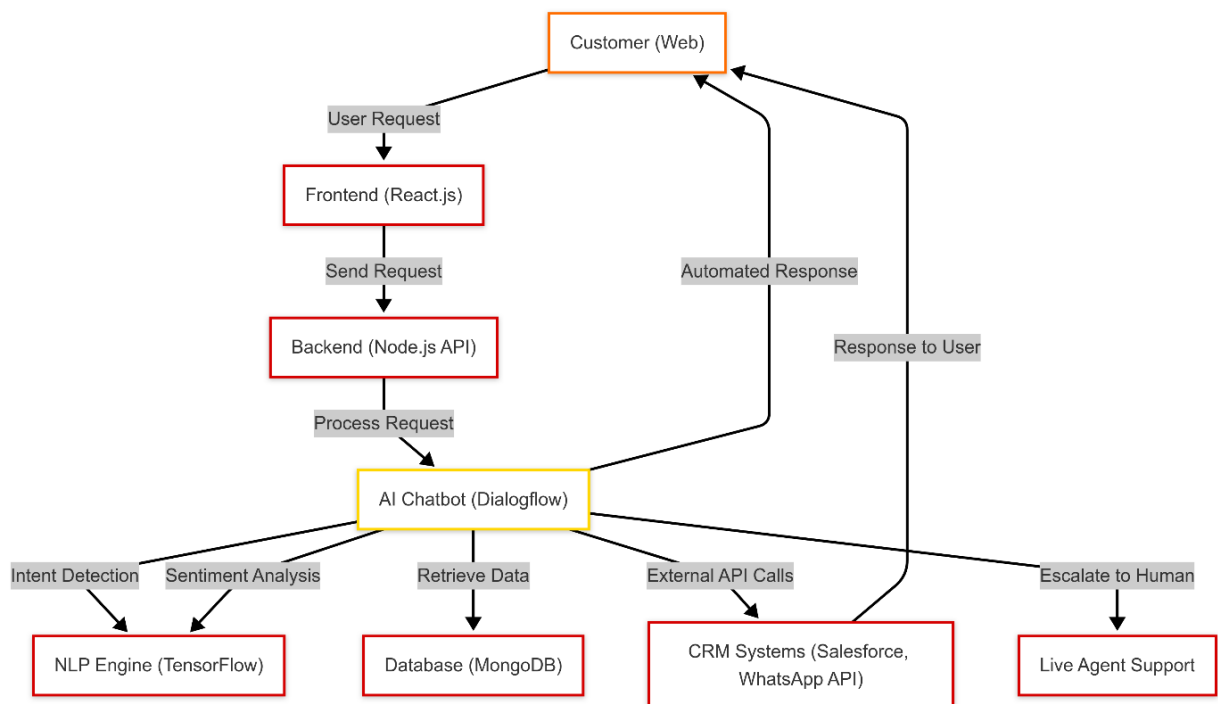
Of Content creating Virtual Assistants at Contact centre

1. Architecture Document

1.1 High-Level Technology Architecture

The technology architecture of AI-powered contact center virtual assistant is a series of layers that communicate with one another in a seamless way to offer intelligent and automated customer care.

1.2 Architecture Diagram / Block Diagram



1.3 Flow of Information Between Component

- 1. The User Interaction:** Customers interact through the web interface (Frontend layer).
- 2. Request Processing:** User requests are passed to the Backend API layer for processing.

3. AI Processing & NLP: AI engine processes and determines intent based on NLP and sentiment analysis.

4. Database Queries: Data pertinent to retrieved from database (e.g., interaction history, problem history).

5. External API Calls: Integration of CRM, communication platform, and other third-party services.

6. Response Generation: AI responds or forwards to human agents as appropriate.

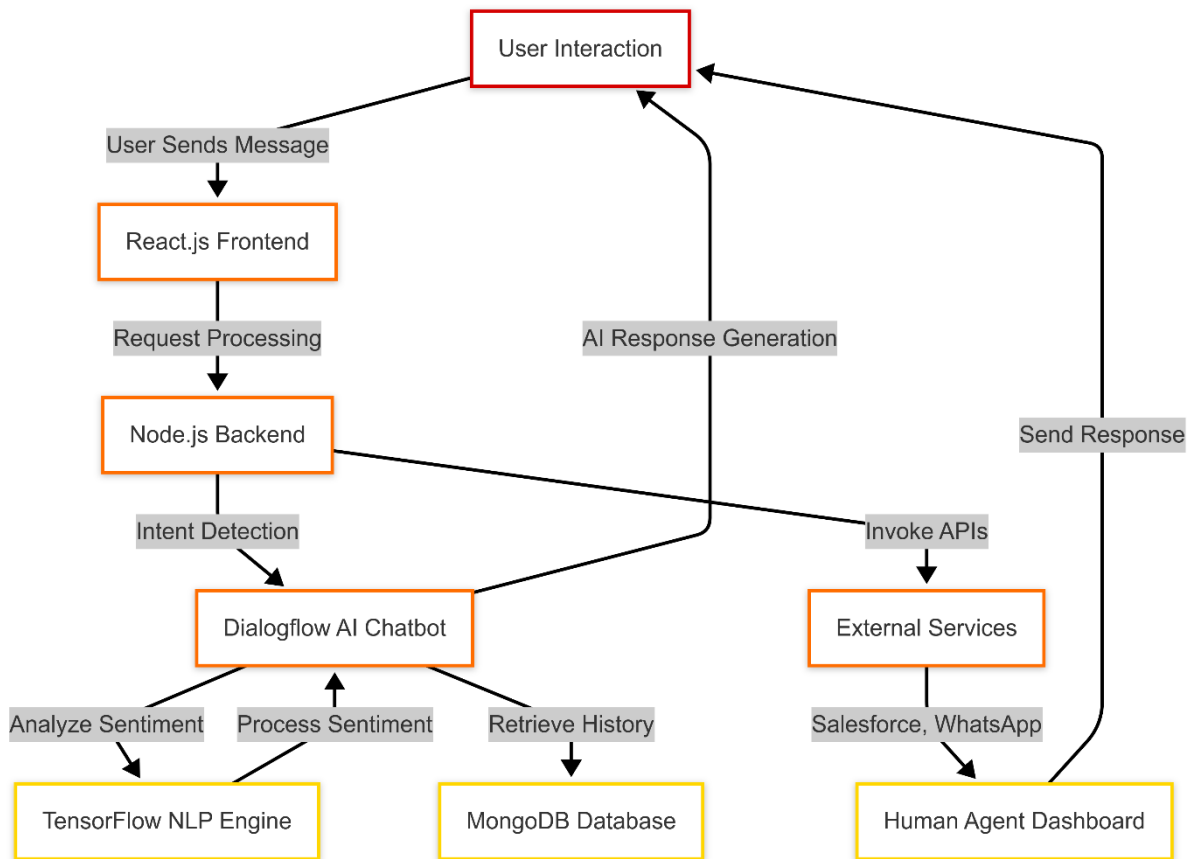
7. UI Updates: Response is returned to the customer on the frontend.

1.4 Blocks Explanation

- Agent and customer user interface (web) - Frontend (UI Layer)
- Processes user requests, runs business logic, and sends data - Backend (API Layer)
- Stores customer conversations, chat history, and AI learning data - Database Layer
- Chatbot automation and predictive analysis run - AI Processing
- Runs sentiment analysis and multi-language processing - NLP Engine
- Integrates with CRM systems - External Integrations

2. Design Document

2.1 Technology Design



2.1.2 Flow of Information

- 1. User Initiates Chat:** Users initiate requests through React.js-based web or mobile apps.
- 2. Request Processing:** Backend (Node.js) processes and receives requests.
- 3. AI Chatbot:** Utilizes Dialogflow for intent detection and routing.
- 4. NLP & Sentiment Analysis:** TensorFlow-based models analyze user sentiments.
- 5. Database Queries:** MongoDB retrieves user history and chat logs.
- 6. API Integrations:** Invokes external systems such as CRM systems or WhatsApp APIs to get a response.
- 7. Response Generation:** Response from AI agent or chatbot is sent back to the customer.

2.1.3 Describing Blocks in Technology Design

- **React.js UI:** Offers a dynamic, responsive user interface to the agents and customers.
- **Node.js Backend:** Handles API calls, request routing, and business logic.
- **MongoDB Database:** Holds chat histories, customer information, and analytics data.
- **AI Chatbot (Dialogflow):** Deals with natural language input and output processing.
- **NLP Engine (TensorFlow):** Tasks with processing sentiment of customers to aid personalization.
- **External APIs:** CRM (Salesforce), communication platforms (WhatsApp, email APIs) for smooth support.